Unlocking Undruggable Targets: Quantum Simulation of Allosteric Signal Propagation

Phase 1 Concept Proposal — Cleveland Clinic Problem Statement
2026 Global Quantum + AI Challenge

1 Problem Framing

More than 85% of disease-causing proteins are undruggable: they have no deep pocket for a small molecule to bind. The only way in is allostery — a hidden, distant site that switches the active site off from afar [1]. Finding these sites classically takes months of molecular-dynamics (MD) sampling of rare motions, and the usual shortcuts are too linear to capture how a signal really travels across a protein.

I treat this as a problem of *quantum signal propagation*. I take the contact network of the apo (unbound) structure as a graph, launch a signal on it, and let it evolve as a continuous-time quantum walk. Its *interference* — coherent, sign-carrying, phase-driven — points to the residues that are dynamically linked to the active site across the fold. The core idea, which I developed and adapted to this problem, is that the *sign* of the time-averaged walk tells constructive interference apart from destructive. Classical diffusion keeps only amplitude ($|U|^2$, always positive) and throws that sign away. Since allostery is exactly a correlated signal sent over a distance, the interference of the fold's vibrational modes is a natural stand-in for it. Everything is predicted *from topology alone*, with no MD.

2 Technical Approach

Paradigm. Gate-based quantum (a continuous-time quantum walk, CTQW), with a fast analytic simulation for development and a hybrid classical–quantum layer on top.

How it works.

1. *Graph.* Apo PDB → residue contact graph (Cα within a cutoff, or fixed-degree k-NN).
2. *Hamiltonian.* The graph Laplacian, or a resonance-coupled version I built where the coupling depends on the log-degree, so residues of similar "frequency" couple more strongly.
3. *Quantum walk.* $U(t) = e^{-iHt}$ spreads the signal over the residues. My core metric is the sign-carrying, time-averaged interference, which I compute in closed form (no time steps):

$$M_{ij} = \sum_a v_a(i)\, v_a(j)\, \mathrm{sinc}(\lambda_a T),$$

   with (λ_a, v_a) the eigenpairs of H. A residue's score adds up its coupling to graph-distant residues (band $d \geq 3$), which isolates the distal signal.
4. *Extra descriptors*, each a method I adapted here: a screened (massive) propagator plus an effective-mass descriptor; an angular packing-balance deviation that flags residues lining a void; and a resonance-overlap read-out, equal to the CTQW transition probability from the active-site superposition.
5. *Putting it together.* An unsupervised *pocket-geometry router* sends each protein to the right detector (proximal, cofactor-anchored → active-site anomaly; distal → interference). A rank *fusion* then blends this physics detector with a learned cross-protein model, because the two carry different, complementary signal.

Adapted (cited) parts. Elastic-network / Gaussian-network baselines for fluctuations [2, 3, 4]; continuous-time quantum walks and quantum simulation of non-local channels [5, 6]; spectral descriptors (heat-kernel signature [7], communicability [8]); and in the learned/QPU layer, gradient boosting [9], symbolic regression [10], quantum boosting as a QUBO [11], and a data-re-uploading variational circuit [12].

3 Feasibility and Resources

Compute. Every metric is an analytic time-average, so I need no MD and no time-stepping. One eigendecomposition per protein is enough ($N \lesssim 10^3$, well under a second on one workstation). I run proteins in parallel, use GPU (CuPy) eigensolves under a flag, and include an eigen-free stochastic estimator (Chebyshev probes) that recovers the score at ∼0.9 correlation for large N.

Quantum hardware. The interference metric maps straight onto gates: I realise the CTQW as an XY-hopping walk in the single-excitation subspace, $H = \sum_{(i,j)} w_{ij}(X_iX_j + Y_iY_j)/2$, Trotterised so each edge is a native `RXX`·`RYY` pair. A real circuit I built in Qiskit [15] reproduces the classical walk; for a 12-node coarse-grained graph it uses 12 qubits, depth ∼120–270, and ∼330–650 two-qubit gates. Under a depolarising model it degrades gently but noticeably, which is why coarse-graining (fewer nodes ⇒ shallower circuits) and error mitigation matter. Circuits export to AWS Braket, Classiq, and IBM.

Data. Public RCSB PDB apo/holo structures; the public Allosteric Database (ASD) [16] is available if I want to grow the training set. No confidential data.

Assumptions / limits. The elastic-network hypothesis (contact topology drives propagation); catalytic domain only; solvent and cofactors dropped except as plain nodes; near-term circuit depth limited by coherence.

4 Expected Impact

A working PoC turns a static apo structure into a ranked map of likely cryptic/allosteric sites — an "allosteric scanner" that sits before expensive screening and can rescue targets once written off as undruggable. My current results already hit the main goal: the fusion predictor scores validated allosteric residues significantly above background on all three benchmarks (Table 1), each with $p < 0.01$ and a bootstrap 95% CI; the single interference detector reaches AUC 0.85 on BCR-ABL1. I also produce a top-5 list for the undruggable transcription factor c-Myc. This lands in the range of physics/dynamics-based methods and below large supervised models trained on hundreds of proteins — a gap I can close by adding ASD structures.

Target (apo→holo)	AUC	95% CI	p
KRAS G12C (4OBE→6OIM)	0.68	$[0.55, 0.79]$	2.8×10^{-3}
BCR-ABL1 (1OPL→5MO4)	0.75	$[0.67, 0.83]$	3.1×10^{-6}
Cardiac Myosin (5TBY→6C1H)	0.73	$[0.63, 0.82]$	4.7×10^{-5}

Table 1: Blind prediction (fusion): apo input, holo-defined pocket as ground truth.

5 Validation Plan

Metric. I separate holo-defined pocket residues (within 5 Å of the drug) from background and score it with AUC, a Mann–Whitney p-value [13], and a bootstrap 95% CI [14]; plus hit@5 and the $N \times N$ connectivity matrix as the main deliverable. **Protocol.** Strictly blind: the unbound structure is the input, the drug-bound one is the ground truth. **Extra checks.** (i) Noise resilience: I model decoherence by sliding a spectral filter from coherent (sinc) to classical (heat). The coherent metric *beats* the classical limit on all three proteins (e.g. myosin 0.71→0.59) with a smooth, monotone drop — so this is robustness and evidence of quantum advantage at once; perturbing the coordinates leaves AUC stable up to 1.5 Å. (ii) Coarse-graining keeps ∼0.90 of the signal at 1/3 size, with eigensolves up to ∼127× cheaper. (iii) Classical baselines. **Success:** significant separation ($p < 0.05$) on every benchmark plus credible quantum-advantage evidence.

6 Hybrid / Cross-Domain Integration

The workflow is hybrid by design: a quantum interference metric (CTQW), classical spectral descriptors, and an AI transfer model, all blended through an unsupervised geometry router. The reason is simple and empirical — the physics detector and the learned model see different, complementary signal, and their rank fusion is steadier than either alone, beating both on the hardest (proximal) case. I specify three QPU-ready forms: the CTQW circuit above; a quantum choice of decision stumps written as a QUBO for annealers or QAOA [11]; and a data-re-uploading variational circuit trained with parameter-shift gradients [12], all exportable to the provided platforms.

7 Team Capability

The originality here is my own theoretical work on signal propagation: the interference, resonance-coupling, screened-propagator, packing-balance, and resonance-overlap methods above are ones I developed and adapted to allosteric prediction. Execution risk is low, because I already have a complete, reproducible pipeline. It produces the connectivity matrices, significance-tested hit lists, real quantum circuits with resource budgets, noise-resilience curves, coarse-graining proofs, and 3D maps that a Phase 2 PoC will formalise on quantum hardware. I bring both this domain-specific method and the software and quantum engineering shown by the working prototype.

References

[1] R. Nussinov, C.-J. Tsai. *Allostery in disease and in drug discovery.* Cell 153, 293 (2013).

[2] B. Erman. *The Gaussian network model.* Biophys. J. 91, 3589 (2006).

[3] C. Chennubhotla, I. Bahar. *Signal propagation in proteins.* PLoS Comput. Biol. 3, e172 (2007).

[4] A. Das *et al. Two-state anisotropic network model.* PLoS Comput. Biol. 10, e1003521 (2014).

[5] E.K. Oh *et al. SVD quantum algorithm for quantum biology.* ACS Phys. Chem. Au 4, 393 (2024).

[6] K. Mitarai, K. Fujii. *Simulating a non-local channel with local channels.* Quantum 5, 388 (2021).

[7] J. Sun, M. Ovsjanikov, L. Guibas. *A concise, provably informative multi-scale signature based on heat diffusion.* Comput. Graph. Forum 28, 1383 (2009).

[8] E. Estrada, N. Hatano. *Communicability in complex networks.* Phys. Rev. E 77, 036111 (2008).

[9] J.H. Friedman. *Greedy function approximation: a gradient boosting machine.* Ann. Statist. 29, 1189 (2001).

[10] J.R. Koza. *Genetic Programming.* MIT Press (1992).

[11] H. Neven *et al. Training a binary classifier with the quantum adiabatic algorithm.* arXiv:0811.0416 (2008).

[12] A. Pérez-Salinas *et al. Data re-uploading for a universal quantum classifier.* Quantum 4, 226 (2020).

[13] H.B. Mann, D.R. Whitney. *On a test of whether one of two random variables is stochastically larger.* Ann. Math. Statist. 18, 50 (1947).

[14] B. Efron. *Bootstrap methods.* Ann. Statist. 7, 1 (1979).

[15] A. Javadi-Abhari *et al. Quantum computing with Qiskit.* arXiv:2405.08810 (2024).

[16] Q. Shen *et al. ASD v3.0: allosteric regulation database.* Nucleic Acids Res. 44, D527 (2016).